Tharun Kumar Marada

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PROFESSIONAL SUMMARY

I finished my MSc in Computer Science in Sweden in June 2026, after two internships where I built machine learning models and Python pipelines. I like the whole path from messy data to something people can use: a model, a dashboard, an agent, a live service. I test my own results hard, because a number that looks good but is wrong is worse than no number. My latest project forecasts Swedish electricity prices every day and publishes its own mistakes. I am looking for a junior role in machine learning, AI or data where I can keep building and keep learning.

TECHNICAL SKILLS

Machine Learning | scikit-learn, XGBoost, forecasting, classification, anomaly detection, walk-forward and grouped validation, SHAP
AI and LLMs | LLM agents (LangGraph), RAG (LangChain, ChromaDB), embeddings (Hugging Face), LLM APIs (Groq, LLaMA 3), AI coding agents
Data | Python, SQL, pandas, NumPy, SQLite, data pipelines, data quality checks, Power BI, Streamlit dashboards
Engineering | Git, CI/CD (GitHub Actions), Docker, pytest, REST APIs (FastAPI, Flask), C++

EDUCATION

MSc Computer Science Blekinge Institute of Technology, Sweden Aug 2024 – Jun 2026

- Thesis: detected AI-generated and paraphrased text with 90.3% accuracy and 0.94 AUC by comparing 9 classifiers built on 21 embedding stability features, using 5,000 samples from the HC3 corpus
- Raised KNN accuracy from 61.6% to 89.3% with the new features, and tested robustness on 25,000 variants generated with LLaMA 3.1

BTech Computer Science and Engineering JNTU, India 2020 – 2024

WORK EXPERIENCE

Machine Learning Intern *Jul 2026 – Sep 2026*
FlyRank.ai

- Reached 0.90 Precision@20 and 0.70 Precision@50 by building a Random Forest model in Python, covering data preparation, feature engineering, training and evaluation
- Kept the reported scores free of data leakage by validating with client-grouped splits, so that no client appeared in both training and test data
- Made results reproducible by building data preparation, transformation and validation as one repeatable workflow

Full Stack Python Intern *Summer 2025*
BIST Technologies

- Kept outputs reliable across Python data and model pipelines by adding automated quality checks at every stage, from incoming data to the final output
- Resolved reliability problems by debugging across the full stack, from data quality to model behaviour to what the user saw

KEY PROJECTS

Electricity Price Forecast Service (Sweden) LIVE CODE

- Cut forecast error by 12% to 15% against a same-hour-yesterday baseline in all 4 Swedish price areas on a 180-day walk-forward test, with bootstrap confidence intervals to separate skill from luck
- Runs by itself every morning in GitHub Actions: collects prices, scores old forecasts and updates a public web page
- Showed the forecast is useful, not just accurate: picking tomorrow's 4 cheapest hours with it cut the price paid by 41% to 48% in southern Sweden compared with no planning
- Kept every change safe with 110 automated tests, a Docker build and a FastAPI service checked in CI on each push

Multi-Agent AI Research Generator LINK

- Produced cited research reports in under 90 seconds by building a 4-agent LangGraph pipeline (planner, searcher, extractor, writer) that draws on 15 to 20 live sources
- Kept each report tied to its sources by splitting fact extraction and writing between two separate agents

Document Question Answering System (RAG) LINK

- Answered questions on a 187-page annual report with cited answers in under 3 seconds by building a RAG pipeline with LangChain, a ChromaDB vector database and LLaMA 3.1

Customer Churn Prediction and Risk Dashboard LINK

- Estimated $1.67M of annual revenue at risk across 7,043 customers by building SQL cohort views and an XGBoost model (0.838 AUC-ROC), and flagged 975 high-risk customers in a live Streamlit dashboard

More on GitHub: football scouting analytics on 1.8M appearances, predictive maintenance and a C++ vehicle ECU simulator.